

The 8 things a Cloud Engine unlocks

A plain-English guide for builders — from someone coming off Rust backend canisters. The one idea: an engine is a **place**, not an app. It's a **dedicated subnet you own**; your canisters run *on* it, unchanged. Same code, same limits — everything here changes the **infrastructure**, never the app. Ranked by what matters most. *(Corrections welcome — docs are still thin.)*

🏰 THE MOAT

1

Encrypted computers (SEV) CONFIDENTIALITY

The chip encrypts the computer's memory. **No one — not even the operator running it — can read your data**, and you can cryptographically attest it.

Why it matters → a guarantee AWS and Google can't make: "not even the host can see it." Needs *all* nodes on SEV — only possible when you pick the nodes, i.e. on an engine.

2

Sovereignty — choose country & operators COMPLIANCE

You pick which countries your computers sit in and which companies run them. On the shared network, placement is automatic.

Why it matters → "EU-only" (or any jurisdiction) becomes enforced by architecture, not a contract clause — the unlock for regulated buyers.

⚙️ OPERATIONAL WINS

3

No lock-in — move freely

PORTABILITY

Once you're on ICP, shift your app to different computers, providers, or countries **without rewriting code or going down** (the subnet "floats" over its nodes; `migrate-id` keeps the canister ID).

Note → freedom *within* ICP — not an AWS→ICP importer.

4

Single-tenant — all yours

PERFORMANCE

No strangers' apps sharing your computers. The whole engine's power and storage is yours — steady, predictable performance under load.

🔧 CONTROL & KNOBS

5

Upgrade control — your approve button

STABILITY

You decide **when the computers' own system software updates** — "install now / schedule / skip." On the shared network it's forced on you. (This is *not* your canister code — that you always control.)

6

Choose how many computers

RESILIENCE

Set the node count / fault tolerance. More computers = more **safety** (copies), not more power or storage.

7

Predictable cost

BUDGETING

Flat per-computer monthly, prepaid — no surprise metered bills. (80% goes to node providers, 20% is burned.)

8

Choose hardware grade

STORAGE & POWER

Pick computer size — small to **dense** (hundreds of GB RAM, tens of TB disk each). Bigger computers = more storage & power per copy.

Storage rule → more storage = **bigger** computers, not **more** of them (more = just copies). One canister maxes at ~500 GB — store more by splitting across several canisters.

THE WHOLE THING IN ONE BREATH

An engine changes *where* your app runs, *who* runs it, *who can see it*, and *how you pay* — never the app itself. #1 and #2 (encrypted + sovereign) are the moat; the rest is how you run it well.